End-to-End Soccer Analysis: Tracking and Event Detection *

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Abstract

We present a fully-automated pipeline for generating soccer statistics directly from raw video, without manual annotations or specialized hardware. Our approach combines contrastive learning for player re-identification and pose-based visual features to improve tracking and event detection. It achieves a HOTA score of 76% on the SoccerNet tracking benchmark, 73.92% mAP on the Soccernet general actions spotting dataset, and 46.2% on the Soccernet ball action spotting dataset. The code can be found on: https://github.com/Streakfull/IDP.

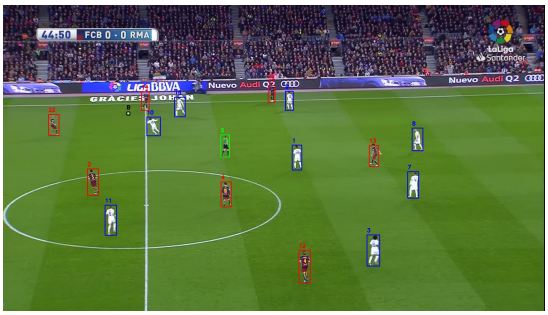

Figure 1. Final visualization of a detected Pass event, showing player detections grouped into three teams along with the identified ball position. Best viewed in color.

1. Introduction

Soccer is among the most widely followed and played sports globally, captivating millions of fans and analysts seeking insights into team and player performances. Traditionally, the computation of soccer statistics has relied on specialized hardware such as high-end multi-camera systems, GPS wearables, and thermal imaging sensors [1]. These setups, while effective, are expensive and impractical to achieve in non-professional settings.

Accurate statistics play a key role in modern soccer, providing teams with a competitive advantage by enabling data-driven tactical decisions [2, 3]. However, automatically extracting these insights from raw video remains a significant challenge.

This paper explores the possibility of generating fully automated soccer statistics directly from raw broadcast or wide-angle videos. This is a highly desirable goal due to its scalability and low operational cost. However, it introduces a set of unique challenges: soccer involves fast-paced, irregular player and ball movements, frequent occlusions, and large-scale camera motions, including pans, zooms, and motion blur. These factors make consistent tracking and identification across long video sequences particularly difficult.

To address this, several interconnected subproblems must be solved: accurate detection of players and the ball, robust multi-object tracking over time, clustering players into teams, and identifying high-level events such as passes, shots, and goals. Each of these components requires careful design to ensure end-to-end reliability. Figure 1 visualizes the final output.

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Our contributions are:

- A contrastive learning approach for improved player re-identification.
- Integration of human pose estimation to enhance visual feature quality.
- Transfer learning to boost performance on the under-represented SoccerNet ball actions dataset.
- A unified pipeline for processing raw soccer video and generating game statistics.

Paper Structure. Section 2 reviews related work. Section 3 details our proposed methodology. Section 4 discusses experiments, results, and some ablation studies. Section 5 provides insights on some limitations and possible next research directions. Section 6 concludes the paper.

2. Related Work

Object detection is the foundation of our tracking pipeline. YOLO [4] revolutionized the field by treating object detection as a single-stage regression task, enabling real-time performance suitable for sports analytics. The YOLO family has evolved rapidly, with the latest being YOLOv11 [5].

Pose estimation enhances player behavior understanding. Toshev and Szegedy [6] first applied deep learning to 2D pose estimation, using a cascaded CNN approach for joint localization. Extensions like YOLO-pose [7] predict both bounding boxes and keypoints in a single pass, achieving state-of-the-art results on COCO [8].

For tracking, we focus on online tracking-by-detection methods, which match object detections to tracklets frame-by-frame using cost-efficient strategies, enabling real-time performance [9, 10, 11, 12].

Long-term tracking faces challenges, such as maintaining identity when players leave and re-enter the scene. Re-identification resolves this by learning consistent visual features, with methods like Habel et al. [13] using contrastive learning with ViT backbones to align player embeddings over time.

The SoccerNet dataset [14] offers a large-scale benchmark for soccer action spotting, with dense annotations for key events across 500+ matches, becoming the standard for soccer video understanding.

For higher-level understanding, Tran et al. [15] introduced a framework combining global context and local entity features. Their Unifying Global and Local (UGL) module uses a CNN backbone and the GLIP model [16] for effective event recognition, with GLIP providing zero-shot object detection guided by text prompts.

3. Methodology

Figure 2 presents an overview of the complete pipeline, illustrating the main processing stages from detection to statistical analysis. The following sections explain each component in detail.

3.1. Player Detection

Accurate player detection is a critical component of our soccer analysis pipeline. We fine-tune the YOLOv11 [5] detector on the SoccerNet dataset [17] to address the specific challenges of sports footage. The loss function combines several components to optimize both classification and localization: the Complete IoU (CIoU) loss [18] for bounding box regression, binary cross-entropy (BCE) loss for classification, and Distributed Focal Loss (DFL) [19] for precise localization. The final loss is given by:

$$\mathcal{L}_{det} = \lambda_{\text{CIoU}}\mathcal{L}_{CIoU} + \lambda_{\text{cls}}\mathcal{L}_{bce} + \lambda_{\text{dfl}}\mathcal{L}_{dfl} \quad (1)$$

where λ_{CIoU}, λ_{cls} and λ_{dfl} are weighting factors set to 7.5, 0.5, 1.5 respectively.

3.2. Ball Detection

Accurate ball detection is critical for recognizing key events such as passes, shots, and goals. We trained a dedicated YOLOv11 [5] model on a custom annotated dataset, built from three 30-second clips of professional matches sourced from YouTube.

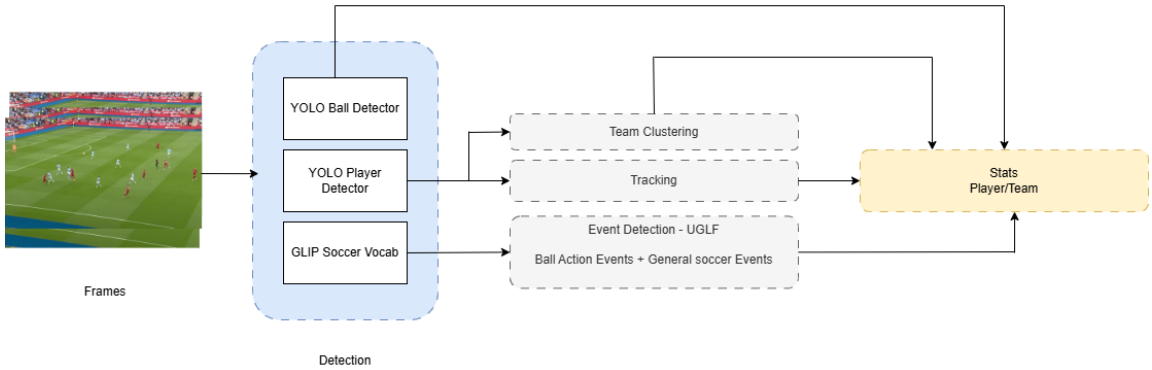

Figure 2. Overview of the full pipeline: frames are processed through object detection (players, ball, actions), followed by team clustering, tracking, and event detection, leading to the generation of player and team statistics.

Using a separate model, rather than a unified player-ball detector, reduces confusion under occlusion and low resolution, significantly improving localization accuracy. The training uses the same loss function as in Section 3.1. Although this adds a second forward pass and slightly increases inference time, the gain in ball detection reliability justifies the trade-off.

3.3. Tracking

Multi-Object Tracking (MOT) enables the continuous tracking of players across frames and is central to our pipeline. As introduced Section 2, we adopt an online tracking-by-detection approach, where detections are linked frame-by-frame using only past information. Player detections (Section 3.1) are matched using spatial and appearance cues to form trajectories, which are essential for downstream tasks such as team clustering in Section 3.3.4 and action spotting in Section 3.4.

3.3.1 Crop features

Features are extracted for each detection based on appearance and 2D pose estimates, as shown in Figure 3, which are then used for matching across frames.

Visual Crop Encoding. We choose a Vision Transformer [20] fine-tuned from CLIP [13], leveraging its pre-training on 400 million image-text pairs to obtain rich, transferable representations

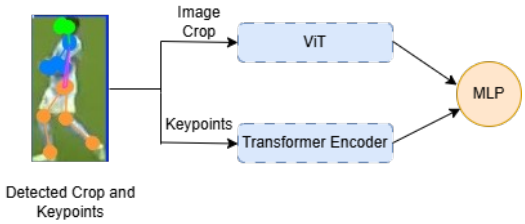

Figure 3. Overview of the feature extractor model: the ViT model encodes the player crop and the transformer encoder encodes the the keypoints.

crucial for capturing fine-grained player appearance.

2D Pose Estimation. Pose estimates serve as another distinguishing feature for player identification. The underlying assumption is that a player's poses remain consistent over short intervals and exhibit recognizable patterns over longer periods. To extract pose information, a pre-trained YOLO-based pose estimator is used [7] to detect key points from player crops. These key points are then fed into a Transformer encoder [21], which learns temporal dependencies and encodes the pose dynamics into a discriminative feature representation.

Fusion. In the final stage of the feature extractor, the appearance $\in \mathbb{R}^{512}$ features and pose-estimate features $\in \mathbb{R}^{512}$ outputs are combined using a simple MLP. This results in the final player feature vector $\mathbf{v} \in \mathbb{R}^{768}$.

Dataset. We opt for the SoccerNet re-

identification dataset [17] rather than the tracking dataset due to two key reasons: (1) the re-identification split spans 500 games across 5 leagues, offering a broader and more diverse set of player crops compared to the tracking split, which consists of only 56 clips from 12 games within a single league [14], and (2) some tracking test clips overlap with training games, leading to potential data leakage. Using the re-identification data ensures improved generalization, which is crucial for real-world application scenarios, the primary objective of this work.

Training. We train the feature extractor model using an infoNCE loss function formulated for images only as in [13].

Let $I_q, I_g \in \mathbb{R}^{n\times H\times W\times C}$ denote batches of query and gallery images. We concatenate and encode them:

$$I = \text{concat}(I_q, I_g), \quad F = \text{Encoder}(I)$$

After ℓ_2 normalization:

$$F = \frac{F}{\|F\|_2}, \quad F_q = F_{1:n}, \quad F_g = F_{n:2n}$$

The pairwise cosine similarity matrices, scaled by a learned temperature t, are:

$$\text{logits}_q = \exp(t) F_q F_g^\top, \quad \text{logits}_g = \text{logits}_q^\top$$

With ground-truth labels labels $= [0, 1, \ldots, n-1]$, the symmetric InfoNCE loss is:

$$\ell_q = \text{CE}(\text{logits}_q, \text{labels}), \quad \ell_g = \text{CE}(\text{logits}_g, \text{labels})$$

$$\mathcal{L}_{infoNCE} = \frac{1}{2}(\ell_q + \ell_g) \tag{2}$$

In addition to training with the InfoNCE loss described in Equation (2), we conduct a separate experiment where the model is trained using the contrastive loss proposed by Chen et al. [22], applied between only two samples. The results of this additional experiment are reported in Section 4.6.2.

3.3.2 EMA Adaptive update

Gran-Henriksen et al. [11] update a track's feature vector whenever a new detection is matched, following Equation (3), where the feature vector v_i^k at frame k is refined using a fixed update weight α

$$v_k^i = \alpha v_i^{k-1} + (1-\alpha) f_i^k \tag{3}$$

.

We propose an adaptive modification: α is dynamically adjusted based on the cosine distance between the current track feature and the new detection. This ensures stronger updates for similar features and smaller updates for dissimilar ones, improving robustness.

Importantly, incorrect matches do not propagate errors, as v_i^k remains unchanged when $\alpha = 1$. The new update rule is:

$$d = 1 - \cos(\theta) = 1 - \frac{\langle \mathbf{v}_i^k, \mathbf{f}_i^k \rangle}{\|\mathbf{v}_i^k\| \|\mathbf{f}_i^k\|} \tag{4}$$

$$\alpha = \begin{cases} 1.0, & \text{if } d > \lambda \\ \frac{d}{\lambda}, & \text{otherwise} \end{cases} \tag{5}$$

$$\mathbf{v}_i^k \leftarrow \alpha \cdot \mathbf{v}_i^{k-1} + (1-\alpha) \cdot \mathbf{f}_i^k, \quad \mathbf{v}_i^k \leftarrow \frac{\mathbf{v}_i^k}{\|\mathbf{v}_i^k\|} \tag{6}$$

where λ controls the maximum considered cosine distance.

3.3.3 Final Matching

Equation (8) [11] defines the association cost as the harmonic mean of EIoU [12] and cosine feature distance, balancing appearance and spatial cues.

$$d_1^{j,k} = 1 - \text{EIoU}(j,k), \quad d_2^{j,k} = 1 - \frac{\langle \mathbf{j}, \mathbf{k} \rangle}{\|\mathbf{j}\| \|\mathbf{k}\|} \tag{7}$$

$$H(j,k) = \frac{2}{\frac{1}{d_1^{j,k}} + \frac{1}{d_2^{j,k}}} \tag{8}$$

Unlike methods relying solely on either feature [10, 11, 23], this approach reduces ID swaps by integrating spatial information.

Hungarian Algorithm. The cost matrix is solved using the Hungarian algorithm [24] for optimal track-detection matching.

Table 1. Small sample of one entity in the created language dictionary

Field	Value
Name	goal
Clean Name	Goal
Definition (Wikipedia)	A goal in soccer is scored when the ball completely crosses the goal line between the goalposts and beneath the crossbar.
Definition (WordNet)	A goal is when the ball passes the goal line, scoring a point for the team.
GPT-4 Definitions	A goal is scored when the ball completely crosses the goal line between the goalposts and under the crossbar. ...
Hierarchy (WordNet)	soccer, scoring, action, event, occurrence

3.3.4 Team Clustering

In this step, we perform team clustering to group players into their respective teams, while non-player classes (e.g., staff or referees) are assigned to an extra cluster. The feature extractor trained in Section 3.1 produces embeddings for each player crop, capturing distinctive characteristics. To improve separability, UMAP [25] dimensionality reduction is first applied, followed by KMeans [26] clustering. Clustering is performed offline using features from the entire video to ensure accurate team assignments by leveraging full context.

3.4. Event Detection

A key component of Li et al. [19] is the construction of a domain-specific language dictionary for soccer entities, enabling the GLIP model [16] to detect soccer-related objects via text prompts without additional training.

We build a 20-class dictionary using a hierarchical strategy, combining GPT-4 generated text [27] with manually curated sources such as Wikipedia. A sample for the "goal" entity is shown in Table 1.

SoccerNet [17] offers two action spotting datasets: one with 17 general events and another with 12 ball-related actions. Our focus is on ball actions, despite limited annotations (only 4 matches).

To overcome data scarcity, we employ a transfer learning approach: pre-training the UGLF framework [15] on 500 games with general action labels, followed by fine-tuning on the ball-action dataset. As shown in Section 4.6.2, this significantly improves ball-action detection while preserving general event performance.

3.5. Automated Stats

Building on the previously introduced components, we design a direct algorithm for extracting per-player and team-level statistics. The procedure is guided by three central questions: (1) What action is currently taking place? (2) Who is in possession of the ball? (3) To which team does the ball belong? The detailed steps of this process are outlined in Algorithm 1.

4. Experiments & Results

All experiments and training procedures were implemented using PyTorch. The models were trained and evaluated on a single NVIDIA RTX 3070 GPU with 10GB of VRAM. This setup was sufficient to train all pipeline components, including detection, tracking, and event-spotting models.

4.1. Tracking

We evaluate tracking performance using HOTA [28] and CLEAR metrics [29], including MOTA, and ID Switches (IDSW).

During inference, we use a maximum matching cost threshold of 0.8 from Equation (8) to associate detections with existing tracks. Additionally,

Algorithm 1: Statistical Analysis Pipeline

```
Input: Event detections ℰ, Frames ℱ,
       Tracklets 𝒯
Initialize counters for possession, passes,
  and events;
foreach e ∈ ℰ do
    Identify player and team from 𝒯;
    if e.label = DRIVE then
        Increment possession counters;
    else if e.label = PASS then
        foreach next DRIVE d within 100
          frames do
            if team(d) = team(e) then
                Mark pass as successful;
                break;
        Increment pass attempts;
    else if e.label ∈ {FREE KICK, SHOT,
      HEADER, BLOCK, CROSS, THROW
      IN} then
        Increment event counters;
Compute possession percentages and pass
  accuracies;
```

crop embeddings are matched only if their cosine distance is below 0.3; otherwise, the detection is treated as a new track.

Table 2 shows that our method outperforms all existing approaches on the SoccerNet test set, achieving the highest scores across HOTA, DetA, and MOTA, while still being very close in the AssA metric. These results demonstrate the effectiveness of our tracker in both detection and association accuracy.

4.2. Event Detection

Following [15], we adopt a 100-frame context window, using frames f_{n-50} to f_{n+50} for each prediction at frame f_n.

Table 3 reports the mAP for general action spotting on the SoccerNet test set, validating the effectiveness of our language dictionary. As defined in [17], the tight (±5s) and loose (±60s) metrics reflect different temporal tolerances for event matching. Our trained UGLF [15] model achieved a precision of 0.8838, recall of 0.8368, accuracy of 0.7538, and an F1 score of 0.8596.

As no published baselines exist for ball action spotting on the 2024 test split covering all 12 actions, Table 4 presents our transfer learning results. High-frequency actions (e.g., Drive, Pass) and visually distinct events (e.g., Goal, Free Kick) are detected with high precision. Overall, we obtain a precision of 0.7489, recall of 0.7704, accuracy of 0.6122, and an F1 score of 0.7595.

Algorithm	**HOTA** ↑	**DetA** ↑	**AssA** ↑	**MOTA** ↑
DeepSORT[10]	36.663	40.022	33.759	33.913
FairMOT[30]	43.911	46.317	41.778	50.698
ByteTrack[31]	47.225	44.489	50.257	31.741
FairMOT-ft[30]	57.882	66.565	50.492	83.565
Deep-EIoU[12]	67.340	70.830	64.110	-
Deep HM-Sort[11]	69.040	71.350	**66.870**	-
Ours	**75.788**	**86.100**	66.700	**86.130**

Table 2. Comparison of tracking algorithms on the SoccerNet test set. Evaluations for other trackers were obtained from [14] and [11]. Our metrics were computed using the official SoccerNet evaluation tools [32].

Method	**Tight**	**Loose**
CALF	-	-
CALF-calib	-	46.80
RMS-Net	28.83	63.49
NetVLAD++	-	-
Zhou et al.	47.05	73.77
Soares et al.	65.07	78.59
E2E-Spot	61.82	74.05
UGLF	62.49	73.98
UGLF(Ours)	62.15	73.92

Table 3. Action spotting mAP on the test set under tight and loose evaluation metrics. Our metrics were computed using the official SoccerNet action spotting evaluation tools [33].

Table 4. Event-wise performance at 1, 3, and 5 frame temporal tolerances.

Class / Tol.	1	3	5
Ball Player Block	9.50	13.47	14.01
Cross	45.03	48.01	49.22
Drive	54.34	65.85	73.89
Free Kick	83.33	83.33	83.33
Goal	100.00	100.00	100.00
Header	39.93	43.73	47.38
High Pass	59.14	63.70	64.52
Out	21.64	35.65	41.82
Pass	69.78	76.95	80.82
Player Successful Tackle	2.20	2.45	2.54
Shot	36.44	41.05	43.69
Throw In	32.59	45.36	53.16
mAP	46.16	51.63	54.53

4.3. Ball Detection

Metric	Fine-tuned	Pre-trained
Precision	77.14	10.53
Recall	84.38	80.00
F1 Score	80.60	18.52
Accuracy	73.47	10.26

Table 5. Comparison of performance metrics between fine-tuned and pre-trained YOLO models on the custom test clip.

For ball detection, we report our results in Table 5, evaluated on a custom 30-second test clip consisting of 990 frames. The YOLO model [5] was fine-tuned specifically for this task. Table 5 also presents the performance of the pre-trained model without task-specific training, highlighting a substantial improvement in precision, F1 score, and accuracy after fine-tuning. Qualitatively, the pre-trained YOLO model failed to detect a large portion of the balls across the frames.

4.4. Team Clustering

#Cls	#UMAP	Precision	Recall	F1
2	3	84.85	84.90	84.87
2	2	84.43	84.48	84.45
2	None	45.74	45.76	45.75
3	3	**86.00**	**86.10**	**86.05**

Table 6. Performance vs. number of classes and UMAP components.

Table 6 presents the performance metrics across different class configurations and UMAP [25] dimensionality settings. Introducing UMAP significantly improves results compared to the baseline without dimensionality reduction, with the best performance achieved using 3 classes and UMAP (F1 score of 86.05). Even with only 2 classes, both 2 and 3 UMAP components yield strong and nearly identical results, highlighting the effectiveness of UMAP in enhancing model performance.

4.5. In the wild

Table 7 shows a final evaluation of the full pipeline and Algorithm 1 on a full broadcasted game. Possession and shots are closely matched. On the other hand, the exact pass and pass accuracy statistics were unavailable for this game; the displayed values represent team averages, and computed values follow similar trends.

4.6. Ablation Studies

4.6.1 Tracking

Table 9 shows that combining keypoint locations with a transformer encoder significantly outperforms using keypoints embeddings (second to last layer output) or not at all. The Vision Transformer (ViT) also outperformed ResNet and OSC backbones, demonstrating superior feature learning.

Fine-tuning the detector proved crucial, with fine-tuned models performing much better than non-fine-tuned ones. Inference at 644p boosted speed (38 FPS vs. 23 FPS) with minimal performance loss, showing efficiency gains are possible

Stat	TeamA		TeamB	
	Comp.	**Actual**	**Comp.**	**Actual**
Possession (%)	43.89	42.6	56.04	57.4
Passes	400	307	430	381
Pass Acc. (%)	73.5	83	78.1	86
Shots	13	14	12	14
Blocks	8	—	4	—
Throw Ins	16	—	14	—
Headers	11	—	8	—
Crosses	8	—	6	—

Table 7. Comparison between computed match statistics and actual statistics obtained from the Premier league and ESPN.

without major accuracy trade-offs.

Training on a re-identification dataset improved generalization to SoccerNet, while InfoNCE loss outperformed contrastive loss for association accuracy. Finally, using only appearance cost reduced ID switches but hurt overall accuracy, highlighting the importance of combining appearance and motion cues.

4.6.2 Event Detection

Training Method	1	3	5
From Scratch	21.02	36.73	46.43
Transfer Learning	46.16	51.63	54.53

Table 8. Comparison of mAP scores at different tolerances for the UGLF model trained from scratch and with transfer learning on the SoccerNet ball actions test set.

Table 8 shows the effectiveness of the transfer learning approach and how utilizing the 500 games from the SoccerNet action spotting dataset improved the ball actions performance by an average of 16% across the three tolerances.

5. Limitations and Future Work

Limitations. While effective, our pipeline currently runs at around 20 FPS and is not real-time. It remains sensitive to occlusions, which can cause occasional ID switches, and longer sequences tend to accumulate tracking errors.

Future Work. We plan to expand our evaluations and training using larger datasets such as SoccerReplay-1988 [34], and other sports footage to improve generalization. SambaMOTR [35] introduced a new sequence modelling approach for MOT that promises better occlusion handling. Testing amateur games and conducting user studies will help assess real-world applicability. Finally, we aim to improve speed and scalability through MediaPipe [36] and explore online team clustering.

Config	HOTA ↑	DetA ↑	AssA ↑	IDSW ↓
Without KP	70.20	85.30	57.88	3382
KP Embedding	74.00	85.69	63.92	2214
ResNet [37]	72.49	85.94	61.15	2437
OSNet [38]	74.40	85.91	64.44	3696
No Fine-tune	64.13	76.78	53.56	3223
644p	68.76	83.07	56.92	2489
W/o Adaptive	74.40	85.83	64.50	2123
SoccerNet Train	75.64	86.13	66.43	2066
Contrastive	73.47	85.90	62.84	2624
IOU Only	58.20	85.69	39.53	9168
Appearance Only	72.17	85.96	60.60	**1922**
Full	**75.79**	**86.10**	**66.70**	1991

Table 9. Tracking performance across different configurations.

6. Conclusion

We presented a fully automated pipeline for soccer analytics that integrates player tracking, re-identification, and event detection. Our method achieves strong performance across multiple benchmarks by combining pose estimation, contrastive learning, and transfer learning. While challenges remain regarding real-time performance and long-sequence tracking, our results highlight the potential of learning-based approaches for structured sports understanding.

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